

CSE208: Digital Logic Design Laboratory

Class 01: Basic Logic Gates & Truth Tables

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Welcome to CSE208

What is Digital Logic?

A digital system processes information using discrete values, most commonly represented by:

0 and 1

Typical interpretation

- 0 → LOW
- 1 → HIGH
- 0 → OFF
- 1 → ON

Applications

- Computers
- Smartphones
- Microprocessors
- Embedded systems

Key Idea

Complex digital systems are built from simple logic gates.

What Will We Do in This Laboratory?

Understand → Design → Build → Simulate → Verify

Logic Gates

- AND
- OR
- NOT
- NAND
- NOR
- XOR
- XNOR

Digital Circuits

- Adders
- Subtractors
- Multiplexers
- Demultiplexers
- Encoders
- Decoders
- Flip-flops

Today's Learning Objectives

By the end of today's laboratory, you should be able to:

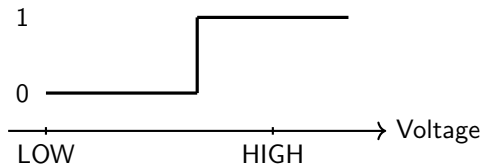
- 1 Explain what a logic gate is.
- 2 Distinguish between LOW (0) and HIGH (1).
- 3 Identify common logic gates.
- 4 Construct truth tables.
- 5 Predict the output of a logic circuit.
- 6 Build basic logic-gate circuits.
- 7 Verify gate behavior using a simulator or hardware.
- 8 Identify common laboratory mistakes.

Digital Logic Levels

Ideal Digital Representation

0 = LOW

1 = HIGH



Important

0 and 1 are logical values. Their actual voltage ranges depend on the underlying logic technology.

What is a Logic Gate?

Definition

A **logic gate** is a digital circuit that performs a Boolean operation on one or more inputs to produce an output.



Example:

$$A, B \longrightarrow AND \longrightarrow Y$$

$$Y = A \cdot B$$

Truth Table

Definition

A **truth table** lists the output of a logic circuit for every possible combination of its inputs.

For two binary inputs A and B :

$$2^2 = 4 \text{ possible combinations}$$

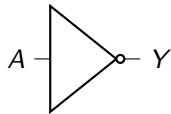
A	B
0	0
0	1
1	0
1	1

General Rule

For n binary inputs:

$$2^n \text{ input combinations}$$

NOT Gate



$$Y = \overline{A}$$

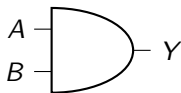
A	Y
0	1
1	0

NOT reverses the input.

$$\text{NOT } 0 = 1$$

$$\text{NOT } 1 = 0$$

AND Gate

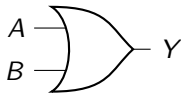


$$Y = A \cdot B$$

<i>A</i>	<i>B</i>	<i>Y</i>
0	0	0
0	1	0
1	0	0
1	1	1

AND = All must be 1

OR Gate



$$Y = A + B$$

<i>A</i>	<i>B</i>	<i>Y</i>
0	0	0
0	1	1
1	0	1
1	1	1

OR = At least one is 1

AND vs OR

AND

$$Y = A \cdot B$$

Output is 1 only for:

11 → 1

OR

$$Y = A + B$$

Output is 0 only for:

00 → 0

Quick Challenge

What are the outputs?

$$1 \cdot 0 = ?$$

$$1 + 0 = ?$$

AND vs OR

AND

$$Y = A \cdot B$$

Output is 1 only for:

11 → 1

OR

$$Y = A + B$$

Output is 0 only for:

00 → 0

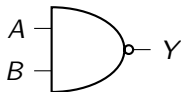
Quick Challenge

What are the outputs?

$$1 \cdot 0 = ?$$

$$1 + 0 = ?$$

NAND Gate

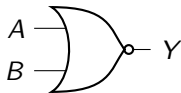


$$Y = \overline{AB}$$

A	B	AND	NAND
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

NAND = NOT(AND)

NOR Gate

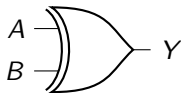


$$Y = \overline{A + B}$$

A	B	OR	NOR
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

NOR = NOT(OR)

XOR Gate

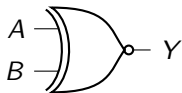


$$Y = A \oplus B$$

<i>A</i>	<i>B</i>	<i>Y</i>
0	0	0
0	1	1
1	0	1
1	1	0

XOR = Inputs are different

XNOR Gate



$$Y = \overline{A \oplus B}$$

<i>A</i>	<i>B</i>	<i>Y</i>
0	0	1
0	1	0
1	0	0
1	1	1

XNOR = Inputs are equal

All Basic Gates at a Glance

Gate	Expression	Output = 1 when...
NOT	$\overline{A}$	$A = 0$
AND	AB	All inputs are 1
OR	$A + B$	At least one input is 1
NAND	$\overline{AB}$	Not all inputs are 1
NOR	$\overline{A + B}$	All inputs are 0
XOR	$A \oplus B$	Inputs are different
XNOR	$\overline{A \oplus B}$	Inputs are equal

AND = All
XOR = Different

OR = Any
XNOR = Same

Combining Gates

Consider:

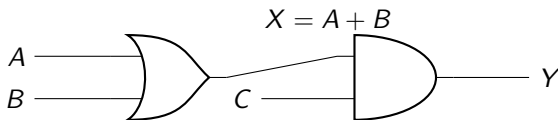
$$Y = (A + B)C$$

We can divide the circuit into two steps:

$$X = A + B$$

then

$$Y = XC$$



Let's Analyze a Circuit

Given:

$$Y = (A + B)C$$

Suppose:

$$A = 0, \quad B = 1, \quad C = 1$$

First:

$$A + B = 0 + 1 = 1$$

Then:

$$Y = 1 \cdot 1$$

Therefore:

$$\boxed{Y = 1}$$

Experiment 01

Verification of Basic Logic Gates

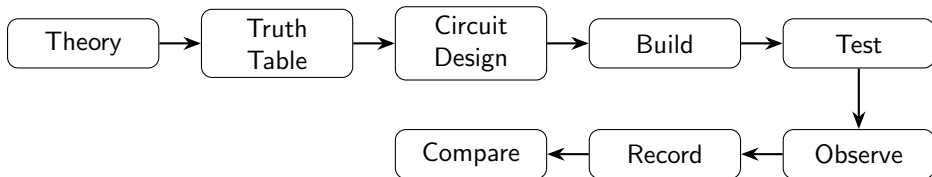
Objective

To verify the truth tables and logical behavior of basic logic gates using a digital simulator or laboratory hardware.

Logic Gates

- AND
- OR
- NOT
- NAND
- NOR
- XOR
- XNOR

Experimental Workflow



Important

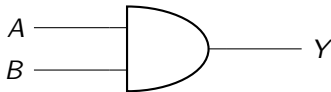
Don't just make the circuit work. Understand why it works.

Circuit Simulation

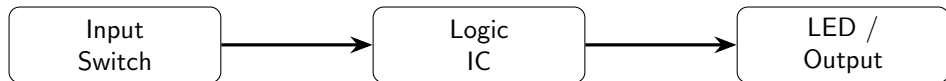
Using a Digital Logic Simulator

For example, in Logisim Evolution:

- 1 Create a new circuit.
- 2 Place input pins/switches.
- 3 Place the required logic gate.
- 4 Place an output/LED.
- 5 Connect the wires.
- 6 Change the input values.
- 7 Observe the output.
- 8 Verify the truth table.



Hardware Experiment



Before powering the circuit, check:

- IC orientation
- VCC connection
- GND connection
- Input connections
- Output connections
- LED polarity
- Loose or misplaced wires

Never leave digital inputs floating!

A floating input may produce:

- Unpredictable output
- Noise
- Intermittent behavior
- Different results from one experiment to another

Golden Rule

Every digital input should have a clearly defined logic state:

0 or 1

Common Laboratory Mistakes

- ❶ **Power:** Is VCC connected correctly?
- ❷ **Ground:** Is GND connected?
- ❸ **IC:** Are you using the correct IC?
- ❹ **Pinout:** Are the correct pins connected?
- ❺ **Inputs:** Are all inputs defined?
- ❻ **Output:** Is the LED/output connected correctly?
- ❼ **Wiring:** Are there loose or misplaced wires?

Troubleshoot systematically.

Quick Quiz #1

What is the output?

$$Y = A \cdot B$$

If:

$$A = 1, \quad B = 0$$

Then:

$$Y = ?$$

Quick Quiz #1

What is the output?

$$Y = A \cdot B$$

If:

$$A = 1, \quad B = 0$$

Then:

$$Y = ?$$

0

Quick Quiz #2

What is the output?

$$Y = A + B$$

If:

$$A = 0, \quad B = 1$$

Then:

$$Y = ?$$

Quick Quiz #2

What is the output?

$$Y = A + B$$

If:

$$A = 0, \quad B = 1$$

Then:

$$Y = ?$$

1

Quick Quiz #3

Which gate produces 1 only when the inputs are different?

- ① AND
- ② OR
- ③ XOR
- ④ XNOR

Quick Quiz #3

Which gate produces 1 only when the inputs are different?

- ① AND
- ② OR
- ③ XOR
- ④ XNOR

XOR

Quick Quiz #4

A circuit has **3 binary inputs**.

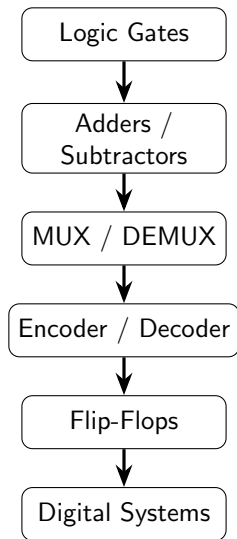
How many rows are required in its complete truth table?

$$2^n$$

Therefore:

$$2^3 = \boxed{8}$$

Why Are Gates Important?



Half Adder & Half Subtractor

We will use today's logic gates to construct arithmetic circuits.

For a half adder:

$$Sum = A \oplus B$$

$$Carry = AB$$

Connection

The XOR and AND gates you learned today will become the building blocks of the half adder.

Today's Takeaways

- ➊ Digital logic uses discrete logical values.
- ➋ A truth table describes every possible input-output combination.
- ➌ For n binary inputs:

$$2^n$$

combinations exist.

- ➍ AND → **All**
- ➎ OR → **Any**
- ➏ XOR → **Different**
- ➐ XNOR → **Same**

You should be able to answer:

- 1 What is a logic gate?
- 2 How many rows does a 4-input truth table have?
- 3 What is the difference between AND and OR?
- 4 Why is XOR different from OR?
- 5 What is a universal gate?
- 6 Why should digital inputs not be left floating?
- 7 Can you build and verify the seven basic gates?

Lab Task — Experiment 01

Verification of Basic Logic Gates

Construct and verify:

Part A

- AND
- OR
- NOT

Part B

- NAND
- NOR

Part C

- XOR
- XNOR

Submit:

- 1 Circuit diagrams
- 2 Truth tables
- 3 Experimental observations
- 4 Verification
- 5 Short conclusion

Digital Logic Starts with One Bit

0 or 1

Gates → Circuits → Digital Systems → Computers

Let's build our first circuit!